TECHNICAL SPECIFICATION DOCUMENT

Import Data Using Transform Maps

Comprehensive solution requirements and architecture specification for importing spreadsheet datasets, configuring transform maps, enforcing security controls, and automating staging workflows.

FUNCTIONAL SOLUTION REQUIREMENTS

1. File Ingestion & Authentication

- Secure ingestion of Microsoft Excel (.xlsx, .xls) and CSV spreadsheets.
- Restrict data upload rights strictly to authenticated platform users.
- Automated generation of staging table schemas upon file upload.

2. Role-Based Data Import Control

- Assign granular import privileges via Role-Based Access Control (RBAC).
- Restrict raw staging data access to authorized operational roles.
- Prevent unauthorized mass data creation or bulk table modifications.

3. Transform Map Management

- Administrative controls to create, update, and retire Transform Maps.
- Establish explicit mapping definitions between staging and target tables.
- Support reusable transform templates for recurring spreadsheet loads.

4. Mapping & Execution Permissions

- **Read:** Inspect staging data rows and field values prior to import.
- **Write:** Modify field relationships, scripts, and coalesce rules.
- **Execute:** Trigger transform processing and target database updates.

DATA VALIDATION & PLATFORM CONTROLS

5. Data Validation & Error Prevention

- Automatic detection and isolation of corrupted or malformed rows.
- Prevent execution errors in staging from impacting target data tables.
- Validate field lengths, data types, and mandatory target attributes.

6. User & Import Set Management

- Administrative oversight to assign and revoke data import privileges.
- Role-based provisioning for managing staging tables and import sets.
- Support multi-user administrative isolation for enterprise environments.

7. Transformation Activity Monitoring

- Comprehensive logging of row insertions, updates, and ignored records.
- Record transformation execution times, session IDs, and failure reasons.
- Provide searchable execution logs for audit and compliance teams.

8. Custom Script & Mapping Requests

- Workflow for operational users to request specialized mapping scripts.
- System administrators review, test, and approve custom transform logic.
- Maintain version control for complex JavaScript transform scripts.

DATA SECURITY & COALESCE RULES

9. Security, Coalesce & Integrity Enforcement

- Enforce strict Access Control Lists (ACLs) across target tables during transformation.
- Mandate Coalesce field configuration (primary key matching) to eliminate duplicate records.
- Apply automated data scrubbing, value normalization, and security policies consistently across every incoming spreadsheet batch.

10. ADMINISTRATIVE DASHBOARD OVERSIGHT

The platform provides a centralized Administrative Dashboard to monitor spreadsheet staging tables, active transform maps, custom mapping scripts, security roles, and real-time execution logs.

TECHNOLOGY STACK OVERVIEW

CATEGORY	TECHNOLOGY ELEMENT
Platform	ServiceNow Platform
Development Tool	ServiceNow Studio
Programming Language	JavaScript (Transform Scripts & Field Mapping)
Database Layer	ServiceNow Database (Staging & Target Tables)
Frontend Interface	ServiceNow UI, Import Set Forms & Lists
Backend Processing	ServiceNow Server-Side Scripting & Import Set Engine
Security Framework	Access Control Lists (ACL) & Data Source Controls
Access Management	Role-Based Access Control (RBAC - `import_admin`)
Authentication	ServiceNow Authentication System
Logging & Monitoring	System Execution & Activity Import Logs
Development Environment	ServiceNow Personal Developer Instance (PDI)

Technical Implementation Summary

The **Import Data Using Transform Maps (Spreadsheets)** project is implemented on the ServiceNow platform using JavaScript for custom field transformation scripts and ServiceNow relational tables for staging and target data management. Access Control Lists (ACLs) and Role-Based Access Control (RBAC) ensure that spreadsheet imports and database updates occur only under authenticated, authorized administrator roles. Built-in system logging delivers complete audit visibility across all data transformation runs.